A Periodic Table of the Finite Simple Groups

Every finite simple group appears exactly once, in its simplest form. Columns are grouped by rank; the vertical axis is the field size q.

TYPE

🟥 Cyclic groups · 🟪 Alternating groups · 🟦 Chevalley groups · 🟩 Steinberg groups · 🟨 Suzuki and Ree groups · 🟪 Sporadic groups

HOW TO READ A CELL

- $\cong A_2(2)$ — other descriptions of the same group
- $A_1(7)$ — the group
- prime factorisation of its order

AN EMPTY CELL

the group is not defined there, or is not simple, or it is isomorphic to one printed elsewhere — every group appears only under its simplest name:

$A_2(1) \to C_3$
$A_1(4), A_1(5) \to A_5\ [A_4(1)]$
$A_1(9) \to A_6\ [A_5(1)]$

$A_3(2) \to A_1(7)$
$A_3(2) \to A_8\ [A_7(1)]$
$^2A_3(2) \to B_2(3)$

$C_n(q)$, q even $\to B_n(q)$
$B_2(2)' \to A_6\ [A_5(1)]$
$G_2(2)' \to {}^2A_2(3)$

$^2G_2(3)' \to A_1(8)$

FAMILIES

$A_n(q) = L_{n+1}(q)$ — Linear
$B_n(q) = O_{2n+1}(q)$ — Orthogonal (odd)
$C_n(q) = S_{2n}(q)$ — Symplectic
$D_n(q) = O^+_{2n}(q)$ — Orthogonal (plus)

$G_2(q)$ — Exceptional G_2
$F_4(q)$ — Exceptional F_4
$E_6(q)$ — Exceptional E_6
$E_7(q)$ — Exceptional E_7

$E_8(q)$ — Exceptional E_8
$^2A_n(q) = U_{n+1}(q)$ — Unitary
$^2D_n(q) = O^-_{2n}(q)$ — Orthogonal (minus)
$^3D_4(q)$ — Triality

$^2E_6(q)$ — Twisted E_6
$^2B_2(q)$ — Suzuki
$^2G_2(q)$ — Ree (small)
$^2F_4(q)$ — Ree (large)

cyclic
rank 1
rank 2
rank 3
rank 4
rank 5
rank 6
rank 7
rank 8
rank 9
rank n
1
2
3
4
5
7
8
9
11
13
16
17
19
23
25
27
29
31
32
q

q = size of the field

Sporadic groups

Mathieu	Mathieu	Mathieu	Mathieu	Mathieu	Janko	Hall–Janko	Janko	Janko	Higman–Sims	McLaughlin	Held	Rudvalis
M_{11}	M_{12}	M_{22}	M_{23}	M_{24}	J_1	J_2	J_3	J_4	HS	McL	He	Ru
$2^4 \cdot 3^2 \cdot 5 \cdot 11$	$2^6 \cdot 3^3 \cdot 5 \cdot 11$	$2^7 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11$	$2^7 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11 \cdot 23$	$2^{10} \cdot 3^3 \cdot 5 \cdot 7 \cdot 11 \cdot 23$	$2^3 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 19$	$2^7 \cdot 3^3 \cdot 5^2 \cdot 7$	$2^7 \cdot 3^5 \cdot 5 \cdot 17 \cdot 19$	$2^{21} \cdot 3^3 \cdot 5 \cdot 7 \cdot 11^3 \cdot 23 \cdot 29 \cdot 31 \cdot 37 \cdot 43$	$2^9 \cdot 3^2 \cdot 5^3 \cdot 7 \cdot 11$	$2^7 \cdot 3^6 \cdot 5^3 \cdot 7 \cdot 11$	$2^{10} \cdot 3^3 \cdot 5^2 \cdot 7^3 \cdot 17$	$2^{14} \cdot 3^3 \cdot 5^3 \cdot 7 \cdot 13 \cdot 29$

Suzuki	O'Nan	Conway	Conway	Conway	Fischer	Fischer	Fischer	Harada–Norton	Lyons	Thompson	Baby Monster	Monster
Suz	O′N	Co_1	Co_2	Co_3	Fi_{22}	Fi_{23}	Fi_{24}'	HN	Ly	Th	B	M
$2^{13} \cdot 3^7 \cdot 5^2 \cdot 7 \cdot 11 \cdot 13$	$2^9 \cdot 3^4 \cdot 5 \cdot 7^3 \cdot 11 \cdot 19 \cdot 31$	$2^{21} \cdot 3^9 \cdot 5^4 \cdot 7^2 \cdot 11 \cdot 13 \cdot 23$	$2^{18} \cdot 3^6 \cdot 5^3 \cdot 7 \cdot 11 \cdot 23$	$2^{10} \cdot 3^7 \cdot 5^3 \cdot 7 \cdot 11 \cdot 23$	$2^{17} \cdot 3^9 \cdot 5^2 \cdot 7 \cdot 11 \cdot 13$	$2^{18} \cdot 3^{13} \cdot 5^2 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 23$	$2^{21} \cdot 3^{16} \cdot 5^2 \cdot 7^3 \cdot 11 \cdot 13 \cdot 17 \cdot 23 \cdot 29$	$2^{14} \cdot 3^6 \cdot 5^6 \cdot 7 \cdot 11 \cdot 19$	$2^8 \cdot 3^7 \cdot 5^6 \cdot 7 \cdot 11 \cdot 31 \cdot 37 \cdot 67$	$2^{15} \cdot 3^{10} \cdot 5^3 \cdot 7^2 \cdot 13 \cdot 19 \cdot 31$	$2^{41} \cdot 3^{13} \cdot 5^6 \cdot 7^2 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23 \cdot 31 \cdot 47$	$2^{46} \cdot 3^{20} \cdot 5^9 \cdot 7^6 \cdot 11^2 \cdot 13^3 \cdot 17 \cdot 19 \cdot 23 \cdot 29 \cdot 31 \cdot 41 \cdot 47 \cdot 59 \cdot 71$

CLASSIFICATION THEOREM

Every finite simple group is cyclic of prime order, alternating of degree n ≥ 5, a group of Lie type, or one of the 26 sporadic groups. This poster shows every one of them, each exactly once.

NOTES

- After the cyclic column the columns are grouped by rank: every family with a simple group of rank 1, then of rank 2, and so on.
- E_8 is the last family to appear, so from rank 9 onwards every block is the same six classical families — hence the ellipsis.
- Row q = 1: over "the field with one element" a group of Lie type degenerates to its Weyl group, and the cell $A_n(1)$ holds the alternating group A_{n+1}. Only type A is simple there, so that row is exactly the alternating groups.
- A cell carrying a parameter shows the generic order formula in place of a prime factorisation, and so does a cell whose factorisation is too long to print.
- $C_n(q) \cong B_n(q)$ for q even, so only the odd-q cells of the C columns are printed.
- $^2B_2(q)$ and $^2F_4(q)$ exist for $q = 2^{2m+1}$ and $^2G_2(q)$ for $q = 3^{2m+1}$ (m ≥ 1); their other rows are empty.
- $^2F_4(2)$ is not simple, but its derived subgroup — the Tits group — is, and is printed in that cell.
- The 26 sporadic groups belong to no family of Lie type; they are shown in their own strip.
- A symbol with a subscript and no field is a group over "F_1": C_p is cyclic of order p and A_m is alternating of degree m. With a field, $A_n(q)$ is linear and $C_n(q)$ symplectic — the argument tells them apart.